

YEAR 7 MATHEMATICS

Chapter 9

The number plane

WORKSHEETS

9

How to use this booklet

1 There is **one worksheet for each lesson**. It has three columns that get harder from left to right: **Easy**, then **Medium**, then **Challenging**.

2 Read the **speech bubble** at the top of each column: it tells you what to do. A new bubble means a new kind of question.

WE DO **Top of each column.** Your teacher works the examples on the board. **Copy every step** into the space.

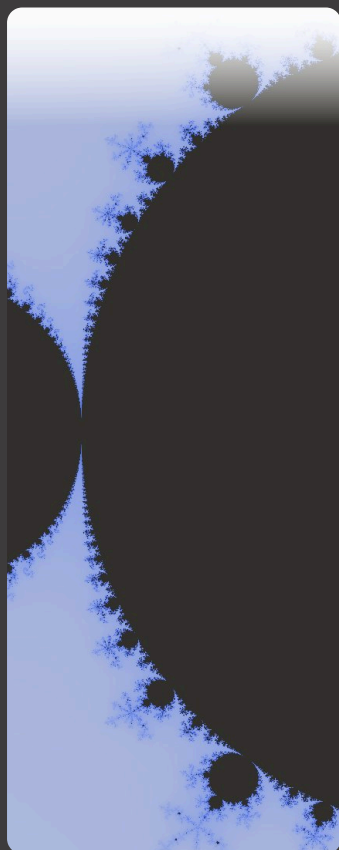
YOU DO **The rest of the column.** Your turn. Column 1: answer on the line. Columns 2 and 3: show your working on the lines.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
9.01	Grid references on maps	3		☐
9.02	Points in the first quadrant	4		☐
9.03	The number plane	5		☐
9.04	Plotting tables of values	6		☐
9.05	Tables of values from rules	7		☐
9.06	Graphing linear equations	8		☐

In this chapter you will

- use grid references on a map
- plot and read points in the first quadrant
- plot and read points in all four quadrants
- plot points from tables and describe patterns
- complete tables of values from rules
- graph linear equations and test whether a point is on a line



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is the whole set, drawn on the complex number plane. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

Easy

Questions 1–16

Use the map. What is at each grid reference? (Letter across, then number up.)

6	Wharf	Fire		Post		Vet		Hotel
5					Club			Beach
4	Farm		Mall				Oval	
3				Gym			Cafe	
2		Shop			Bus			Zoo
1	Pool		Bank			Park		
	A	B	C	D	E	F	G	H

WE DO

With your teacher: copy the working.

E1 A1

E2 A4

YOU DO

Your turn.

1 D6

Post

2 H2

3 H5

4 H6

5 D3

6 F6

7 F1

8 C4

9 A6

10 E2

11 B6

12 C1

13 G3

14 G4

15 B2

16 E5

Medium

Questions 17–26

Use the map above. Write the grid reference, then describe the move from the first place.

WE DO

With your teacher: copy the working.

E1 Bus to Fire

E2 Oval to Shop

YOU DO

Your turn.

17 Park to Mall

Mall is at C4

3 left, 3 up

18 Farm to Post

Post is at D6

___ right, ___ up

19 Fire to Bank

Bank is at C___

___ right, ___ down

20 Hotel to Gym

21 Shop to Club

22 Mall to Wharf

23 Post to Zoo

24 Bank to Cafe

25 Gym to Vet

26 Club to Pool

Challenging

Questions 27–32

Follow the directions on the map. Write where you end up.

WE DO

With your teacher: copy the working.

E1 Start at the Bank. Go 5 right and 5 up. Where are you?

E2 Start at the Shop. Go 5 right and 1 up. Where are you?

YOU DO

Your turn.

27 Start at the Farm. Go 5 right and 3 down. Where are you?

start: A4

across +5 → F; up/down -3 → 1

F___: the Park

28 Start at the Beach. Go 5 left and 4 down. Where are you?

start: H___

across ___ → C; up/down ___ → ___

C___: the Bank

29 Start at the Vet. Go 3 left and 2 down. Where are you?

30 Start at the Bank. Go 2 right and 1 up. Where are you?

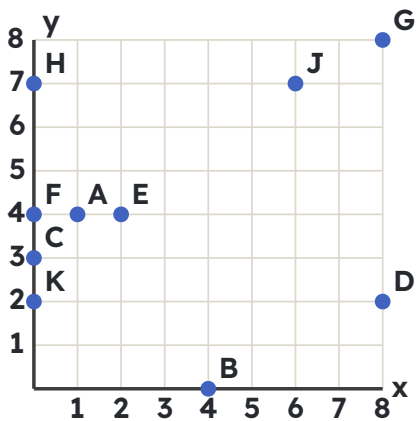
31 Start at the Farm. Go 3 right and 1 down. Where are you?

32 Start at the Beach. Go 7 left and 4 down. Where are you?

Easy

Questions 1–8

Write the coordinates of each point: across (x) first, then up (y).



WE DO With your teacher: copy the working.

E1 point A

E2 point B

YOU DO Your turn.

1 point C (0, 3)

2 point D _____

3 point E _____

4 point F _____

5 point G _____

6 point H _____

7 point J _____

8 point K _____

Medium

Questions 9–18

The two points are on the same horizontal or vertical line. How far apart are they?

WE DO With your teacher: copy the working.

E1 (9, 6) and (9, 0)

E2 (8, 2) and (8, 8)

YOU DO Your turn.

9 (6, 4) and (9, 4)

same y, so count across: $9 - 6$

= 3 units

10 (2, 3) and (9, 3)

same y, so count across: $9 - 2$

= ____ units

11 (7, 8) and (0, 8)

same y, so count across: ____ - ____

= ____ units

12 (7, 8) and (7, 3)

13 (6, 3) and (6, 10)

14 (7, 2) and (7, 4)

15 (7, 3) and (7, 8)

16 (1, 9) and (8, 9)

17 (4, 3) and (2, 3)

18 (3, 10) and (3, 8)

Challenging

Questions 19–25

Three corners of a rectangle are given. Find the fourth corner. Sketch it first.

WE DO With your teacher: copy the working.

E1 (2, 4), (9, 4), (9, 9)

E2 (4, 1), (7, 1), (7, 9)

YOU DO Your turn.

19 (0, 4), (9, 4), (9, 9)

same x as (0, 4): $x = 0$

same y as (9, 9): $y = 9$

fourth corner (____, ____)

20 (1, 1), (8, 1), (8, 5)

same x as (____, ____): $x =$ ____

same y as (____, ____): $y =$ ____

fourth corner (____, ____)

21 (1, 0), (5, 0), (5, 7)

22 (4, 4), (7, 4), (7, 7)

23 (2, 1), (5, 1), (5, 7)

24 (3, 4), (6, 4), (6, 6)

25 (4, 3), (6, 3), (6, 5)

Easy

Questions 1–29

Which quadrant is the point in (1st, 2nd, 3rd or 4th)? Or is it on an axis?

WE DO With your teacher: copy the working.

E1 $(-7, 4)$

E2 $(4, -3)$

YOU DO Your turn.

1 $(-1, 5)$

2nd

2 $(5, 3)$

3 $(8, -7)$

4 $(-4, -2)$

5 $(-7, -1)$

6 $(5, 8)$

7 $(-2, -1)$

8 $(-8, 9)$

9 $(-3, -8)$

10 $(-5, 0)$

11 $(2, -3)$

12 $(2, 2)$

13 $(2, 4)$

14 $(8, -6)$

15 $(-5, 2)$

16 $(8, -2)$

17 $(-2, -3)$

18 $(2, 1)$

19 $(6, 6)$

20 $(-6, -8)$

21 $(3, 7)$

22 $(6, -7)$

23 $(-7, 6)$

24 $(0, 9)$

25 $(-5, -2)$

26 $(-4, 4)$

27 $(4, -8)$

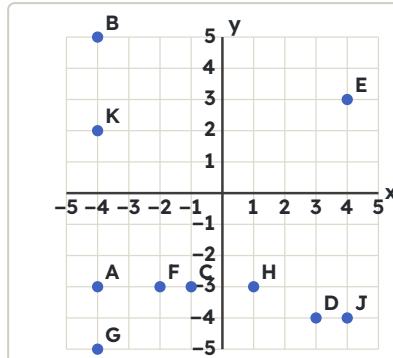
28 $(-5, -8)$

29 $(1, 9)$

Medium

Questions 30–37

Use the number plane. Write the coordinates of each point, then its quadrant.



WE DO With your teacher: copy the working.

E1 point A

E2 point B

YOU DO Your turn.

30 point C

$(-1, -3)$

3rd quadrant

31 point D

$(3, -4)$

4th quadrant

32 point E

$(_, _)$

st quadrant

33 point F

34 point G

35 point H

36 point J

37 point K

Challenging

Questions 38–44

Reflect the point in an axis. Describe what happens to the signs.

WE DO With your teacher: copy the working.

E1 Reflect $(4, 5)$ in the y-axis.

E2 Reflect $(2, -7)$ in the y-axis.

YOU DO Your turn.

38 Reflect $(-2, -4)$ in the x-axis.

x-axis: the y-coordinate changes sign

$(-2, -4) \rightarrow (-2, 4)$

quadrant: 3rd $\rightarrow$ 2nd

39 Reflect $(-8, 4)$ in the x-axis.

x-axis: _____

$(_, _) \rightarrow (_, _)$

quadrant: 2nd $\rightarrow$ 1st

40 Reflect $(5, -7)$ in the y-axis.

41 Reflect $(3, -6)$ in the x-axis.

42 Reflect $(6, 7)$ in the y-axis.

43 Reflect $(-4, 8)$ in the y-axis.

44 Reflect $(-7, 4)$ in the y-axis.

Easy

Questions 1–30

Write each column of the table as a point (x, y).

WE DO With your teacher: copy the working.

E1 $x = 0, y = 5$

E2 $x = 6, y = 5$

YOU DO Your turn.

1 $x = -1, y = 7$ (-1, 7)

2 $x = -2, y = -4$

3 $x = 2, y = 0$

4 $x = -2, y = 0$

5 $x = 4, y = 6$

6 $x = 1, y = 12$

7 $x = 0, y = 11$

8 $x = 4, y = -4$

9 $x = -2, y = -2$

10 $x = 0, y = -4$

11 $x = 4, y = 2$

12 $x = 6, y = 9$

13 $x = 3, y = 0$

14 $x = 6, y = 10$

15 $x = 5, y = 12$

16 $x = 3, y = 12$

17 $x = -3, y = 1$

18 $x = 0, y = 9$

19 $x = 2, y = -3$

20 $x = 5, y = 9$

21 $x = 0, y = 8$

22 $x = -3, y = 5$

23 $x = 3, y = 3$

24 $x = 5, y = 1$

25 $x = 2, y = 4$

26 $x = -1, y = 0$

27 $x = 5, y = 4$

28 $x = -2, y = -1$

29 $x = 0, y = 12$

30 $x = 5, y = 3$

Medium

Questions 31–40

The points follow a pattern. What happens to y each time x goes up by 1? Write the next point.

WE DO With your teacher: copy the working.

E1 (1, 8), (2, 13), (3, 18)

E2 (0, 2), (1, -1), (2, -4)

YOU DO Your turn.

31 (1, 3), (2, 6), (3, 9)

y goes up by 3

next: (4, 12)

32 (0, 8), (1, 9), (2, 10)

y goes up by 1

next: (____, ____)

33 (2, 10), (3, 12), (4, 14)

y goes up by ____

next: (____, ____)

34 (0, 6), (1, 10), (2, 14)

35 (0, 7), (1, 6), (2, 5)

36 (2, 10), (3, 14), (4, 18)

37 (2, 11), (3, 13), (4, 15)

38 (3, 16), (4, 21), (5, 26)

39 (2, -5), (3, -8), (4, -11)

40 (0, 8), (1, 7), (2, 6)

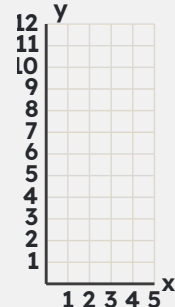
Challenging

Questions 41–41

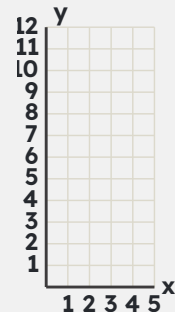
Plot the points on the grid, join them, and describe the pattern.

WE DO With your teacher: copy the working.

E1 Plot (0, 1), (1, 4), (2, 7), (3, 10).

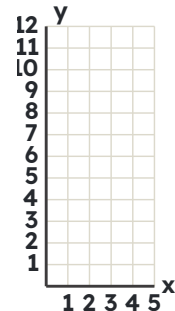


E2 Plot (0, 2), (1, 3), (2, 4), (3, 5).



YOU DO Your turn.

41 Plot (0, 3), (1, 4), (2, 5), (3, 6).



the points make a straight line

each time x goes up ____, y goes up ____

Easy

Questions 1–28

Substitute the value of x into the rule to find y .



WE DO

With your teacher: copy the working.

E1 $y = 5x + 6, x = 7$

E2 $y = 2x - 5, x = 1$

YOU DO

Your turn.

1 $y = 5x + 3, x = 6$

33

2 $y = 4x - 4, x = 6$

3 $y = 3x - 4, x = 2$

4 $y = 4x - 3, x = 6$

5 $y = 3x + 8, x = 7$

6 $y = 3x + 9, x = 1$

7 $y = 4x + 7, x = 4$

8 $y = 4x - 2, x = 6$

9 $y = 6x - 2, x = 0$

10 $y = 3x - 1, x = 6$

11 $y = 3x - 4, x = 5$

12 $y = 6x + 7, x = 8$

13 $y = 4x + 6, x = 5$

14 $y = 2x + 5, x = 4$

15 $y = 3x + 8, x = 5$

16 $y = 6x + 5, x = 2$

17 $y = 4x - 2, x = 0$

18 $y = 2x - 2, x = 1$

19 $y = 2x - 1, x = 6$

20 $y = 6x - 5, x = 6$

21 $y = 2x + 7, x = 4$

22 $y = 6x + 8, x = 5$

23 $y = 3x - 1, x = 7$

24 $y = 5x + 8, x = 1$

25 $y = 4x + 8, x = 1$

26 $y = 6x + 4, x = 7$

27 $y = 3x + 1, x = 3$

28 $y = 2x - 3, x = 3$

Medium

Questions 29–37

Complete the table for $x = 0, 1, 2, 3$. Show the substitution for the first one.



WE DO

With your teacher: copy the working.

E1 $y = 4x + 8$

E2 $y = 5x$

YOU DO

Your turn.

29 $y = 3x + 1$

$x = 0: y = 3 \times 0 + 1 = 1$

$y: 1, 4, 7, 10$

30 $y = 5x + 7$

$x = 0: y = 5 \times 0 + 7 = 7$

$y: _, _, _, _$

31 $y = 3x + 6$

$x = _: y = _ \times _ + _ = _$

$y: _, _, _, _$

32 $y = 2x + 7$

33 $y = 2x + 3$

34 $y = 5x + 4$

35 $y = x$

36 $y = 5x + 9$

37 $y = 3x + 8$

Challenging

Questions 38–43

Find the rule for the table: $y = (\text{how much } y \text{ goes up})x + (\text{y when } x = 0)$.



WE DO

With your teacher: copy the working.

E1 $x: 0, 1, 2, 3$
 $y: 0, 6, 12, 18$

E2 $x: 0, 1, 2, 3$
 $y: 5, 10, 15, 20$

YOU DO

Your turn.

38 $x: 0, 1, 2, 3$
 $y: 6, 12, 18, 24$

y goes up by 6, so $6x$

when $x = 0, y = 6$

$y = _x + _$

39 $x: 0, 1, 2, 3$
 $y: 7, 12, 17, 22$

y goes up by $_$, so $_x$

when $x = _, y = _$

$y = _x + _$

40 $x: 0, 1, 2, 3$
 $y: 8, 9, 10, 11$

41 $x: 0, 1, 2, 3$
 $y: 9, 10, 11, 12$

42 $x: 0, 1, 2, 3$
 $y: 7, 11, 15, 19$

43 $x: 0, 1, 2, 3$
 $y: 0, 5, 10, 15$

Easy

Questions 1–29

Does the point lie on the line? Substitute x and check y . Write yes or no.

WE DO With your teacher: copy the working.

E1 $(3, 7); y = 2x + 1$

E2 $(-2, 1); y = x + 3$

YOU DO Your turn.

1 $(-1, 2); y = x + 1$

no

2 $(-3, -8); y = 2x - 2$

3 $(-2, 2); y = x + 4$

4 $(-1, 0); y = 2x + 2$

5 $(-1, 1); y = 2x + 1$

6 $(5, 26); y = 4x + 3$

7 $(2, 10); y = 4x + 2$

8 $(4, 8); y = x + 2$

9 $(3, 6); y = 2x$

10 $(2, 10); y = 3x + 1$

11 $(-3, -1); y = 2x + 5$

12 $(5, 13); y = 2x + 3$

13 $(2, 1); y = 2x - 3$

14 $(1, 10); y = 4x + 5$

15 $(-2, -5); y = 3x + 1$

16 $(2, 13); y = 4x + 5$

17 $(4, 18); y = 3x + 3$

18 $(4, 6); y = 2x - 2$

19 $(5, 7); y = x + 2$

20 $(-2, -3); y = 2x - 1$

21 $(-1, -4); y = 3x - 1$

22 $(5, 20); y = 4x - 1$

23 $(3, 2); y = x - 1$

24 $(2, 5); y = 4x - 3$

25 $(3, 10); y = 2x + 4$

26 $(-3, -6); y = 4x + 3$

27 $(1, 6); y = 2x + 4$

28 $(4, 3); y = x - 1$

29 $(1, 7); y = 4x + 3$

Medium

Questions 30–39

Make a table of values for $x = -1, 0, 1$. Write the points to plot.

WE DO With your teacher: copy the working.

E1 $y = 2x - 2$

E2 $y = -3x + 1$

YOU DO Your turn.

30 $y = x + 1$

$y: 0, 1, 2$

points: $(-1, 0), (0, 1), (1, 2)$

31 $y = 3x + 2$

$y: -1, 2, 5$

points: $(_, _), (_, _), (_, _)$

32 $y = -x + 2$

$y: _, _, _$

points: $(_, _), (_, _), (_, _)$

33 $y = x - 2$

34 $y = 3x + 4$

35 $y = x - 1$

36 $y = x + 4$

37 $y = -2x + 2$

38 $y = -2x + 1$

39 $y = -2x + 4$

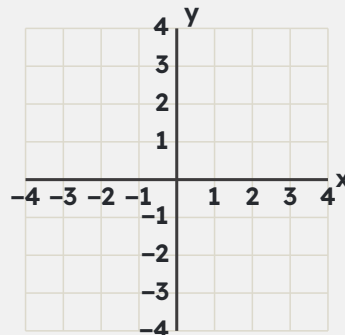
Challenging

Questions 40–40

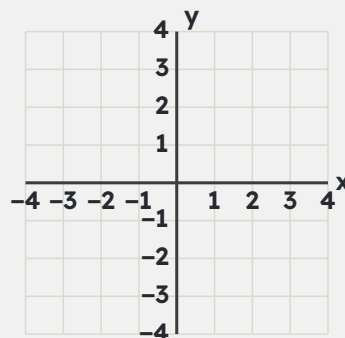
Graph the line on the number plane. Then write where it cuts the y -axis.

WE DO With your teacher: copy the working.

E1 Graph $y = 2x - 1$.

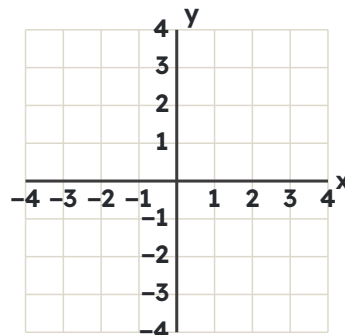


E2 Graph $y = -2x - 2$.



YOU DO Your turn.

40 Graph $y = 2x + 3$.



points: $(-1, 1), (0, 3), (1, 5)$

cuts the y -axis at $(_, _)$